8.1 Unit description

Context approach

This unit covers thermal energy, nuclear decay, oscillations, and astrophysics and cosmology. The unit may be taught using either a concept approach or a context approach. This section of the specification is presented in a format for teachers who wish to use the context approach. The context approach begins with the consideration of an application that draws on many different areas of physics, and then the laws, theories and models of physics that apply to this application are studied. The context approach for this unit uses two different contexts: building design and cosmology.

Concept approach

This unit is presented in a different format on page 37 for teachers who wish to use a concept approach. The concept approach begins with a study of the laws, theories and models of physics and then explores their practical applications. The concept approach is split into four topics: thermal energy, nuclear decay, oscillations, and astrophysics and cosmology.

How Science Works

How Science Works – Appendix 3 should be integrated with the teaching and learning of this unit.

It is expected that students will be given opportunities to use spreadsheets and computer models to analyse and present data, and make predictions while studying this unit.

The word 'investigate' indicates where students should develop their practical skills for *How Science Works*, numbers 1–6 as detailed in *Appendix 3*.

Students should communicate the outcomes of their investigations using appropriate scientific, technical and mathematical language, conventions and symbols.

Applications of physics should be studied using a range of contemporary contexts that relate to this unit.